

CivicConnect — PED Baseline

Project — Milestone 1

SEN381 — Software Engineering 381

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1. Problem & Business Need

Right now, the organisation that CivicConnect is being built for handles service requests — things like facility faults, damaged equipment, security concerns, IT support issues, maintenance requests and lost property — through a mix of email, phone calls, WhatsApp messages, spreadsheets and paper records, with no single system actually owning the process from start to finish. Because nothing tracks a request end-to-end, requests get duplicated or lost, requesters have no real visibility into what's happening with their issue, it's often unclear which staff member is actually responsible, and any reporting that does happen is manual, inconsistent and hard to trust.

So the real value CivicConnect needs to deliver isn't just "a website for logging issues" — it's a controlled, traceable record that follows every request from submission through to closure. That means requesters can actually trust their request won't disappear, staff get a workload they can manage and prioritise, and management gets evidence it can rely on for outstanding, overdue and resolved work. This ties directly back to the problems raised above: duplication, lost visibility, weak accountability and reporting nobody can audit.

Because we're a 2-person team delivering this within a single semester, we've deliberately bounded the business need: the platform has to improve visibility and accountability, but it can't create a technical, operational or financial burden we can't realistically sustain. That framing is behind most of the scope and constraint decisions made in Sections 3 and 4.

2. Stakeholder Analysis

We identified four primary stakeholder groups, plus the project team itself, as having a direct interest in CivicConnect's requirements. Influence and interest are rated qualitatively (Low/Medium/High) so it's clear whose expectations should win out when trade-offs come up.

Stakeholder	Needs	Influence	Interest	Conflicts / Competing Expectations
Community Requester (staff/community member submitting a request)	A simple way to log a fault/issue; visibility of status; confidence the request will not be lost.	Low	High	Wants fast resolution and minimal effort to submit; may resist being restricted to a single digital channel instead of a quick phone call or WhatsApp message.

Stakeholder	Needs	Influence	Interest	Conflicts / Competing Expectations
Front-line Staff (technicians/support staff actioning requests)	Clear, prioritised view of assigned work; enough detail to act without chasing the requester; a low-friction way to update status.	Medium	High	Wants flexibility in how/when status is updated; conflicts with management's need for consistent, timely status accuracy for reporting.
Management / Oversight	Reliable, auditable visibility of open, overdue and resolved work; defensible reporting; accountability for who did what and when.	High	High	Wants comprehensive reporting and strict status discipline; conflicts with staff's preference for lighter-weight, less frequent updates.
Organisation (business owner of CivicConnect)	A sustainable, low-cost solution that measurably reduces lost/duplicated/mishandled requests without a large operational or financial burden.	High	High	Wants broad capability (e.g. every channel supported); conflicts with the cost/schedule constraint that favours a smaller, controlled, high-quality solution over a large unfinished one.
Project Team (student engineers)	A scope that is achievable, defensible and demonstrable within the SEN381 milestone schedule and individual skill level.	Medium	High	Wants to defer high-risk/high-effort features (e.g. third-party channel integration); conflicts with organisation's and requester's preference for familiar channels such as WhatsApp/telephone.

2.1 Hardest Conflict to Reconcile

The hardest conflict to reconcile is between the Requester's preference for familiar, low-effort channels like WhatsApp or a phone call, and the whole point of the project, which is a single controlled record. If requesters were still allowed to submit through informal channels, CivicConnect would basically just become a sixth channel added on top of the five that are already causing the problem, instead of actually fixing anything. We resolved this by requiring all requests to go through the controlled platform (see FR-001–FR-002), though we do acknowledge in the Risk Register (Section 7) that this creates an adoption risk that needs to be monitored rather than assumed away.

The requirements below come directly out of this analysis: FR-001 to FR-005 cover the Requester's need for a low-friction, visible way to submit and track requests; FR-006 to FR-011 cover Staff's need for a manageable, prioritised and accountable workload; FR-012 to FR-015 cover Management's need for auditable oversight.

3. Scope Baseline

3.1 In Scope

- Digital submission of a service request with a controlled category, description, location and optional attachment (FR-001, FR-002).
- Requester-facing status tracking and personal request history (FR-003, FR-004).
- Basic in-platform feedback/notification on status change (FR-005).

- Staff triage: viewing, searching, filtering, assigning and updating requests through controlled status transitions, with recorded actions/comments (FR-006–FR-011).
- Management oversight: status summary, overdue identification and filtered reporting views (FR-012–FR-014).
- Role-based access control for three roles: Requester, Staff, Management (FR-015).

3.2 Out of Scope (for this project)

- Native mobile applications — a responsive web platform is the committed interface (supports NFR-008).
- Direct integration with external channels the organisation currently uses (WhatsApp Business API, telephony/IVR, email-to-ticket parsing).
- Automated SLA-based escalation workflows beyond a basic overdue flag (FR-013).
- Multi-language support and payment/billing functionality — not part of the CivicConnect business need.

3.3 Deferred / Future Scope

- Exportable analytics/reporting (CSV/PDF) beyond in-platform filtered views.
- Push/email notification delivery beyond in-platform status visibility.
- Automated SLA escalation and reminder workflows.
- Integration with external ticketing/ERP systems or a formal WhatsApp/telephony intake channel.
- Multi-organisation / multi-tenant support.

3.4 Defended Exclusion: External Channel Integration (WhatsApp/Telephony)

This exclusion is deliberate, and we're defending it rather than just quietly leaving it out. Integrating WhatsApp or telephony intake would mean an extra external API dependency, ongoing third-party service costs, and handling the security and privacy of messaging data outside our own controlled environment — that's more than a 2-person team can realistically design, secure, test and defend properly within the SEN381 milestone schedule (Master Brief Section 4, Section 3.1 on scope discipline). More importantly, CivicConnect's whole value comes from being a single controlled record of the request lifecycle, so bringing back the exact channel fragmentation the platform is meant to replace would work against that rather than support it. Requiring submission through the controlled platform is therefore both a scope-discipline call and a way of reinforcing the actual business need. We haven't thrown channel integration away, though — it's kept as future scope (Section 3.3), since the underlying need for low-effort submission is still valid and worth revisiting once the core workflow is proven.

4. Constraints

Constraint	Description	Engineering Implication for CivicConnect
Scope	Committed scope (Section 3) must be baselined and controlled; changes require the change-management process (Master Brief Section 14).	Additional features (e.g. channel integration) cannot be silently absorbed mid-project; any addition must go through formal impact analysis.
Schedule	Four formal milestones must be completed within the SEN381 delivery period with a 2-student team.	Favours a smaller, fully controlled feature set (Section 3.1) over a larger, partially-finished one, consistent with Master Brief Section 3.1.
Cost / Resources	Free or low-cost services are preferred; operational cost beyond the educational context must be identified.	Hosting/database/notification service selection (Milestone 2) must document free-tier limits, directly shaping NFR-002 and NFR-009.
Quality	Quality attributes must be defined now and supported by measurable evidence later, not asserted.	Each NFR (Section 5.2) is written with a measurable acceptance criterion so later milestones can produce test evidence against it rather than an unsupported claim.
Security	Security is a lifecycle-wide responsibility from requirements through to deployment, not a final add-on.	Role-based access (FR-015) and confidentiality/auditability (NFR-004, NFR-005) are baselined at M1 so later architecture and construction milestones must design and test against them from the outset.

4.1 Constraint Interaction / Trade-off

The Schedule constraint interacts directly with Quality. Realistically, a 2-person team can't deliver exhaustive automated test coverage across every single requirement within the SEN381 schedule, on top of everything else required for M1–M4 (architecture, construction, security, deployment). Our response is to prioritise by risk: the highest-risk, highest-value workflows — request submission (FR-001–FR-002), controlled status transitions (FR-010) and role-based access (FR-015, NFR-004) — get the deepest verification effort in later milestones, while lower-priority, Should-level capabilities like management summary filtering (FR-012–FR-014) get proportionately less. We're recording that trade-off here explicitly rather than letting it happen quietly later, in line with Master Brief Section 18, which says schedule pressure shouldn't silently reduce testing or security.

5. Requirements & Acceptance Criteria

The requirements below follow the identifier scheme FR-0XX for functional and NFR-0XX for non-functional, each with a source, a MoSCoW-aligned priority and a testable acceptance criterion. This lets each one trace end-to-end into the RTM and get verified later with actual evidence instead of an unsupported claim.

5.1 Functional Requirements

ID	Requirement	Source	Priority	Acceptance Criteria
FR-001	Requester can submit a new service request providing category, description, location and optional attachment.	Requester capability (Master Brief Section 3)	Must	On valid submission the system creates a request with a unique reference number, timestamp and status "Submitted"; the requester sees an on-screen confirmation.

ID	Requirement	Source	Priority	Acceptance Criteria
FR-002	Requester must select a request category from a controlled list (Facility Fault, Damaged Equipment, Security Concern, IT Support, Maintenance, Lost Property, Other).	Requester capability	Must	Submission is blocked until a category is selected; selecting "Other" requires a short free-text description.
FR-003	Requester can view the current status of a specific submitted request.	Requester capability	Must	Given a valid reference number, the system returns one of: Submitted, Assigned, In Progress, Resolved, Closed.
FR-004	Requester can view a history/list of all requests they have previously submitted.	Requester capability	Must	List displays all of the requester's own requests, newest first, each showing reference number, category and current status.
FR-005	Requester receives feedback when a request's status changes (accepted, rejected, updated, completed).	Requester capability	Should	Within one status-check or notification cycle after a staff update, the requester-visible status reflects the change; the previous state is not shown as current.
FR-006	Authorised staff can view the list of service requests relevant to their role.	Staff capability	Must	A logged-in staff user sees only requests they are authorised to view, not administrative or unrelated organisational data.
FR-007	Staff can search, filter and sort requests by category, status, date and requester.	Staff capability	Must	Applying any single filter or a combination reduces the visible list to matching requests only, with no loss of data integrity.
FR-008	Staff can view full request detail, including description, requester reference, category, location and prior actions.	Staff capability	Must	Opening a request shows all recorded fields and the full action/comment history for that request.
FR-009	Staff can assign themselves or another authorised staff member to a request.	Staff capability	Must	An unassigned request can be claimed or assigned; the request record then shows an owning staff member and an "Assigned" status.
FR-010	Staff can update a request's status only through controlled transitions (Submitted → Assigned → In Progress → Resolved → Closed).	Staff capability	Must	An attempt to move a request to a non-adjacent or invalid state (e.g. Submitted → Closed) is rejected with an explanatory message.
FR-011	Staff can record actions, comments or resolution notes against a request.	Staff capability	Must	A saved comment/action is permanently attached to the request record with author and timestamp, and is visible in the request's history.
FR-012	Management can view a summary of service activity by status (open, assigned, in progress, resolved, closed).	Management capability	Should	A summary view shows a count of requests in each status, matching the underlying request data at the time of viewing.
FR-013	Management can identify overdue requests using a defined time threshold per category.	Management capability	Should	A request exceeding its category's defined threshold while not Resolved/Closed is flagged as overdue in the management view.

ID	Requirement	Source	Priority	Acceptance Criteria
FR-014	Management can view/filter requests by category, status or date range for reporting purposes.	Management capability	Should	Applying a management filter returns a correct, matching subset of requests usable for a basic activity report.
FR-015	The system authenticates users and enforces role-based access so each account can only use the capabilities of its assigned role (Requester, Staff, Management).	Cross-cutting / security implication of Section 3 roles	Must	A user assigned the Requester role cannot access staff or management-only screens or actions, verified by direct URL/action attempts as well as normal navigation.

5.2 Non-Functional Requirements

ID	Category	Requirement	Measurable Acceptance Criteria	Priority
NFR-001	Usability	A first-time requester can submit a request without prior training.	Task-based usability check: an untrained test user completes a submission in ≤3 minutes with no assistance.	Should
NFR-002	Availability	The platform is available during normal organisational operating hours, consistent with a free/low-cost hosting constraint.	Uptime monitored over a defined observation window; target ≥99% during business hours, with any outage logged and explained.	Should
NFR-003	Performance	Core list and status pages respond within an acceptable time under expected usage for an organisation of this size.	Measured page-load/response time ≤2 seconds under normal (non-peak) load in the deployed environment.	Should
NFR-004	Security – Confidentiality	Request and account data are only visible to authenticated users authorised for their role; credentials are never stored in plaintext.	Verified by role-based access tests (see FR-015) and inspection of the credential storage mechanism (hashed, not plaintext).	Must
NFR-005	Auditability	Every status change and assignment is recorded with the acting user, timestamp and prior/new value.	A sample of status changes can be traced end-to-end in the system's history log with no missing actor or timestamp.	Must
NFR-006	Data Integrity	A request's category and status must always be one of the controlled, predefined values.	Attempting to set a category or status outside the controlled list is rejected at the point of entry.	Must
NFR-007	Maintainability	Requirements, risks and decisions are traceable through consistent unique identifiers to support later design, test and change control.	Every requirement/risk/decision entered into the PED/RTM carries a unique ID that is used consistently in later milestone artefacts.	Should
NFR-008	Accessibility / Responsiveness	Requester-facing screens are usable on both desktop and common mobile browsers, reflecting that community requesters use varied devices.	Key requester screens (submit, status, history) render and remain usable at common mobile and desktop viewport widths.	Should
NFR-009	Cost	The platform operates within free or low-cost service tiers appropriate for an educational deployment.	Selected hosting/services documented against their free-tier limits; any cost beyond free-tier is identified and recorded in the PED.	Must

5.3 Traceability Note

Every requirement above traces back to a named stakeholder source (Section 2) and a scope decision (Section 3): Requester needs map to FR-001–FR-005, Staff needs to FR-006–FR-011, Management needs to FR-012–FR-014, and the cross-cutting Security constraint (Section 4) to FR-015 and NFR-004–NFR-006. This mapping feeds straight into the Initial RTM, which turns source → requirement → acceptance-criteria traceability into a live, controlled artefact.

6. Initial Requirements Traceability Matrix (RTM)

The Requirements Traceability Matrix establishes an unbroken, auditable link from original stakeholder needs and project constraints through to verification and release. In accordance with Master Project Brief Section 11.1, the full lifecycle schema spans: Stakeholder/Source → Requirement ID → Design/ADR → Issue/PR → Implementation Module → Test ID → Release Evidence.

For Milestone 1, source-to-requirement and acceptance criteria bindings are baselined. Downstream columns (Design, PR, Code, Test, Release) are established as controlled structural placeholders to prevent requirement orphanhood during Milestones 2–4.

6.1 Baseline Traceability Matrix

Req ID	Source	Requirement Description	Acceptance Criteria Summary	Planned Module	Test Plan ID
FR-001	Requester	Digital submission with category, description, location, attachment	Unique ticket ID generated, status set to "Submitted", receipt displayed	RequestIntakeService	TC-FR001-01
FR-002	Requester	Controlled category selection + mandatory "Other" description	Dropdown selection validated; arbitrary strings rejected	CategoryFilter	TC-FR002-01
FR-003	Requester	Lookup request status via reference number	Valid ID returns current lifecycle state; invalid returns 404	RequestQueryHandler	TC-FR003-01
FR-004	Requester	View personal historical request list	Authenticated user sees own submitted items in reverse chronological order	RequesterDashboard	TC-FR004-01
FR-005	Requester	State transition feedback & notification	In-app status reflects change within 1 notification/polling cycle	NotificationService	TC-FR005-01
FR-006	Staff	Filtered operational request access by role	Staff views only role-authorised departmental tickets	StaffTriageGateway	TC-FR006-01
FR-007	Staff	Multi-criteria search, filter, and sort	Filtering by category/status/date yields exact matching subset	RequestSearchEngine	TC-FR007-01
FR-008	Staff	Complete ticket inspection with audit history	UI displays metadata, description, attachments, and full action log	TicketDetailView	TC-FR008-01

Req ID	Source	Requirement Description	Acceptance Criteria Summary	Planned Module	Test Plan ID
FR-009	Staff	Work ticket claiming and reassignment	Owner field updated; ticket transitions to "Assigned" status	AssignmentManager	TC-FR009-01
FR-010	Staff	Strict finite-state transition enforcement	Invalid transitions (e.g. Submitted → Closed) rejected with 400 Bad Request	RequestWorkflowEngine	TC-FR010-01
FR-011	Staff	Immutable resolution notes and staff comments	Comments appended with immutable actor ID and UTC timestamp	AuditCommentService	TC-FR011-01
FR-012	Management	Aggregate service activity dashboard	Visual counters accurately match underlying DB record counts	ManagementSummaryView	TC-FR012-01
FR-013	Management	Automated overdue flag based on category SLA	Tickets active past threshold time display visual warning flag	SLACalculatorService	TC-FR013-01
FR-014	Management	Date-range and departmental reporting filters	Exportable/renderable filtered subset without query degradation	ReportingQueryService	TC-FR014-01
FR-015	Security	Role-Based Access Control (Requester, Staff, Admin)	Direct URL manipulation to elevated endpoints returns 403 Forbidden	RBACSecurityFilter	TC-FR015-01
NFR-001	Usability	First-time requester submission ≤ 3 min	Usability test passing ≥ 90% test pool on first attempt without assistance	IntakeFormUI	TC-NFR001-01
NFR-002	Availability	99% operational uptime during business hours	Monthly downtime ≤ 1% within 08:00–18:00 window under free tier	CloudInfraMonitoring	TC-NFR002-01
NFR-003	Performance	P95 server response ≤ 2 seconds on non-peak loads	Automated load tests confirm P95 latency under target ceiling	APIEndpoints	TC-NFR003-01
NFR-004	Security	Hashed credentials; role-isolated data boundaries	Zero plaintext passwords in DB; BCrypt/Argon2 hashing verified	AuthIdentityEngine	TC-NFR004-01
NFR-005	Auditability	Full audit capture of state changes and actors	100% of state change events write an append-only log record	SystemAuditLogger	TC-NFR005-01
NFR-006	Data Integrity	Input whitelist validation against predefined enums	Enums reject arbitrary strings at API controller boundary	DTOValidationLayer	TC-NFR006-01
NFR-007	Maintainability	Systematic traceability across all lifecycle artefacts	All commits, PRs, and test cases reference canonical requirement IDs	EngineeringRepo	TC-NFR007-01
NFR-008	Responsiveness	Cross-viewport layout parity (Mobile & Desktop)	Clean rendering across 375px to 1920px without horizontal scroll breakage	WebClientFrontend	TC-NFR008-01
NFR-009	Cost	Deployment constrained strictly within free tiers	Zero infrastructure cost incurred across educational lifecycle	HostingEnvironment	TC-NFR009-01

7. Risk Register & Management Strategy

In accordance with Master Project Brief Section 12, the Risk Register is a live control artefact that tracks technical, schedule, security, and process uncertainties. Risk Exposure is scored as: Exposure = Probability (1–3) × Impact (1–3).

7.1 Live Project Risk Register

Risk ID	Risk Description & Root Cause	P × I	Score	Mitigation Strategy (Preventative)	Contingency Plan (Reactive)	Owner
RSK-001	Free-tier cloud exhaustion / cold-starts: Provider (Render/Neon) caps compute hours or spins down mid-defence.	3 × 3	9 (High)	Benchmark quotas in M2; configure synthetic keep-alive pings during assessment window.	Maintain containerised docker-compose setup capable of running 100% offline.	Daluvuyo Magagane
RSK-002	User intake resistance: Community resists web form, preferring WhatsApp/calls.	2 × 3	6 (High)	Design frictionless mobile web intake (NFR-001) requiring no pre-registration.	Introduce staff "proxy logging" mode for telephonic ticket capture.	Thato Goqo
RSK-003	Team skill gap in state-engine architecture: Delays during M3 integration.	1 × 3	3 (Med)	Execute technical spikes in M2; choose mature, well-documented frameworks.	Scope down optional features (e.g. defer SLA escalation FR-013) to protect core triage.	Daluvuyo Magagane
RSK-004	Merge conflicts & config drift: Concurrent PRs overwrite shared DB models.	2 × 2	4 (Med)	Strict branch isolation per ticket; enforce mandatory peer PR approval before merge.	Technical Lead assists with interactive git rebase before PR merging.	Daluvuyo Magagane
RSK-005	Secret leakage to GitHub: Accidental commit of API keys or connection strings.	3 × 2	6 (High)	Configure global .gitignore in M1; implement pre-commit secret detection hooks.	Immediate credential revocation, git history rewrite (git filter-repo), and re-issuance.	Daluvuyo Magagane
RSK-006	Unverified AI code hallucinations: LLM generates deprecated/insecure code.	3 × 1	3 (Med)	Enforce AI Usage Register (Section 11); peer-review all AI snippets against unit tests.	Reject unverified PRs; revert and rewrite code using official framework documentation.	Thato Goqo
RSK-007	Scope creep into external integrations: Mid-project demands for WhatsApp API.	2 × 1	2 (Low)	Explicit scope baseline (PED Section 3.2); defend exclusion formally in Section 3.4.	Enforce Change Management process; reject unapproved changes via Impact Analysis.	Thato Goqo
RSK-008	Superficial test assertions: High raw code coverage masking untested edge cases.	1 × 2	2 (Low)	Define risk-based testing standards in M1/M2 targeting state transition validity.	Assessor defence will probe assertions; reject PRs with assertion-free tests during review.	Thato Goqo

7.2 Defence of Highest-Priority Risk: RSK-001 (Cloud Free-Tier Exhaustion & Cold Starts)

Context & Risk

Bound by constraint NFR-009 (zero educational budget), the team relies on public cloud free tiers (e.g. Render, Railway, Neon). These services enforce 30–60 second cold-start delays after inactivity and hard monthly CPU/egress caps, which could lead to platform failure during evaluation.

Mitigation

In M1, performance metrics (NFR-002, NFR-003) are explicitly baselined for non-peak conditions. In M2, all infrastructure configuration will be decoupled from business logic via environment variables. As a definitive contingency, an offline multi-container docker-compose profile will be maintained to guarantee live demonstration readiness independent of cloud availability.

8. Forward Engineering Considerations Register

In accordance with Milestone 1 Brief Section 4, the engineering baseline systematically anticipates 6 downstream lifecycle decisions before committing to code:

1. Secret Management & CI Injection

Why It Matters Now: Hardcoded secrets committed early become permanent git liabilities, violating Master Brief Section 9.

Downstream Impact: Shapes M2 hosting configuration and M3 CI/CD automation.

Missing Information: Final hosting provider selection (e.g. Render vs. Azure App Service).

Risk of Ignoring: Credential leaks requiring destructive git history rewrites.

2. Relational vs. Document Persistence

Why It Matters Now: Requirements FR-008, FR-010, and FR-011 demand strict relational integrity and append-only audit logs.

Downstream Impact: Dictates M2 schema normalisation and ORM selection.

Missing Information: Benchmark data on join latency vs JSON field flexibility.

Risk of Ignoring: Adopting schemaless NoSQL allowing corrupt state mutations.

3. Automated CI Gates & Tooling Quotas

Why It Matters Now: Quality claims require automated verification evidence (Master Brief Section 15).

Downstream Impact: Determines M3 GitHub Actions workflow runners and build speeds.

Missing Information: Test suite execution duration on free GitHub runner minutes.

Risk of Ignoring: Slow pipelines prompting developers to bypass branch rules.

4. Stateless Auth & Session Scalability

Why It Matters Now: FR-015 mandates RBAC across three roles, dictating backend auth architecture.

Downstream Impact: Governs M2 API design, route guarding, and token management.

Missing Information: JWT token expiry complexity vs stateful session memory overhead.

Risk of Ignoring: Stateful sessions failing on ephemeral auto-scaling instances.

5. Cold-Start Mitigation & Health Probes

Why It Matters Now: Free-tier cloud instances hibernate, threatening NFR-003 response targets.

Downstream Impact: Shapes M2 hosting choices and M4 operational readiness.

Missing Information: Exact sleep timeouts across candidate cloud providers.

Risk of Ignoring: Demo timeouts during M3/M4 assessor evaluations.

6. Append-Only Audit Logging Architecture

Why It Matters Now: NFR-005 requires immutable state change histories. Retrofitting auditing is costly.

Downstream Impact: Dictates M2 domain events and database trigger design.

Missing Information: Event sourcing model vs dedicated audit log table.

Risk of Ignoring: Destructive SQL updates that destroy audit compliance.

9. Engineering Decision Log (EDL)

In accordance with Master Project Brief Section 13, all significant architectural and process decisions are formally recorded:

DEC-001: Adoption of Responsive Web Architecture over Native Mobile Development

Context & Constraints: CivicConnect must serve community requesters on mobile devices and staff on desktops under a 2-student team constraint with zero budget (NFR-009).

Alternatives Considered: (1) Cross-platform mobile app (Flutter/React Native) + Web Admin. (2) Native mobile apps. (3) Single responsive Web App.

Decision & Rationale: Option 3. Developing separate codebases triples UI work and fragments test suites under M3. A single responsive web interface fulfills NFR-008 cleanly.

Trade-offs & Risks: Sacrifices native OS push alerts; requires disciplined CSS design to prevent mobile UX friction.

DEC-002: Structured Branching Strategy & Mandatory Peer PR Review

Context & Constraints: Master Brief Section 9 mandates protected main and peer review before merge, with zero self-merges.

Alternatives Considered: (1) Direct commits to main. (2) Feature branches with mandatory review by the other team member. (3) Protected main requiring the standard two-reviewer model.

Decision & Rationale: Option 2, adapted for a 2-person team. With only one non-author team member available, every substantive change requires that member's explicit review and approval before merge (1-

of-1 non-author sign-off) rather than the two-reviewer model assumed for 3-person teams. This is recorded here as a deliberate, defended departure from the Master Brief's default reviewer count, since it is not achievable with a 2-person roster — see the note under Section 10.2.

Trade-offs & Risks: A single reviewer per PR gives weaker review coverage than a two-reviewer model; mitigated by both members being accountable for the full codebase and by keeping PRs small and frequent.

DEC-003: Formal Deferment of Technology Stack & DB Selection to Milestone 2

Context & Constraints: Milestone 1 boundary rules forbid premature tech-stack lock-in prior to baselined requirements.

Alternatives Considered: (1) Prematurely select familiar tools (ASP.NET vs Express). (2) Formally defer selection to M2.

Decision & Rationale: Option 2. Architecture must be driven by baselined requirements (FR-010, NFR-003, NFR-004) rather than developer habit.

Downstream Consequence: M2 will deliver comprehensive weighted trade-off matrices for candidate stacks via ADRs.

10. GitHub Governance, Configuration Management & Team Working Agreement

10.1 Repository Layout

```
civicconnect-root/
├── .github/
│   ├── workflows/           # CI/CD pipeline automation (Milestone 3)
│   ├── PULL_REQUEST_TEMPLATE.md # Mandatory PR review checklist & quality gate
│   └── CODEOWNERS           # Repository ownership assignments
├── docs/                   # Controlled Engineering Artefacts
│   ├── PED/               # Project Engineering Document releases (v1.0 - v4.0)
│   ├── requirements/      # Baselined RIM and specifications
│   ├── architecture/     # UML models, wireframes, data designs (Milestone 2)
│   ├── decisions/        # Architecture Decision Records (ADRs)
│   ├── risk/             # Live Risk Register and tracking logs
│   ├── change/           # Formal change requests & impact analyses
│   ├── quality/          # Test strategies, metrics, defect register
│   └── security/         # Threat models, vulnerability scans
├── deployment/           # Environment configurations, Dockerfiles
├── src/                  # Application source code (Milestone 3 construction)
├── tests/               # Automated unit, integration, and regression suites
├── .gitignore            # Enterprise ignore rules for secrets and build artefacts
└── README.md             # Engineering documentation hub
```

10.2 Branch Protection & Pull Request Protocol

Protected main Branch: Direct commits are blocked. All code must arrive via feature branches: feat/FR-XXX-name or fix/RSK-XXX-name.

Mandatory Peer Approval: Every PR must receive explicit approval from the other (non-author) team member before merge. Self-approval is rejected. Note: the Master Project Brief's default reviewer model

assumes a 3-student team and asks for 2 non-author approvals; with a 2-person team only one non-author reviewer exists, so this project applies a defended 1-of-1 non-author approval rule instead (see DEC-002).

Structured Review Lifecycle: Every PR executes: Review Comment → Author Response → Correction → Re-review → Approval → Merge.

Automated Secret Protection: Pre-commit hooks and repository secret scanning automatically block commits containing API tokens, passwords, or connection strings.

11. Responsible Artificial Intelligence (AI) Engineering Register

In accordance with Master Project Brief Section 10, all material AI usage is recorded with human verification applied:

Date	User	Tool	Task	Verification & Testing	Decision	Issues / Corrections
2026-09-02	Thato Goqo	Claude 3.5	Drafting initial FR requirements	Reconciled against CivicConnect Master Brief; removed 10 enterprise ITIL capabilities.	Modified	AI suggested multi-tenant billing; rejected as out-of-scope.
2026-09-04	Daluvuyo Magagane	ChatGPT-4o	Structuring RTM schema & risk tables	Verified schema against Master Brief Section 11.1; refined risks for free-tier hosting limits.	Modified	AI proposed Kubernetes; rejected as violation of NFR-009.
2026-09-06	Daluvuyo Magagane	Copilot	Generating polyglot .gitignore	Manually checked against security secret policy; added rules for .env.local and certificates.	Accepted (Edits)	Missed environment override files; appended manually.
2026-09-07	Daluvuyo Magagane	Claude 3.5	Drafting state transition regex logic	Tested against FR-010 finite state rules; identified transition bypass vulnerability.	Modified	AI allowed Submitted → Resolved jump; fixed to enforce sequence.

12. PED v1.0 Baseline Sign-Off & Gate Decision

12.1 Governance Gate Verification Checklist

Governance Check	Verification Standard Applied	Status	Lead
Scope Baseline Reviewed	In/out/deferred boundaries explicit; external channel exclusion defended	PASSED	Thato Goqo
Requirements Baseline Checked	15 FRs & 9 NFRs defined with unique IDs, MoSCoW priorities & testable criteria	PASSED	Thato Goqo
Traceability Established	100% of requirements mapped to source, test plan, and planned module	PASSED	Daluvuyo Magagane
Risk Register Completed	8 project risks scored ($P \times I$); highest risk (RSK-001) fully defended	PASSED	Daluvuyo Magagane
Forward Engineering Analysed	6 lifecycle concerns (Auth, DB, CI, Secrets, Cold-start, Audit) documented	PASSED	Daluvuyo Magagane

Governance Check	Verification Standard Applied	Status	Lead
GitHub Governance Operational	Protected main and peer-review PR ruleset (adapted for a 2-person team, DEC-002), and repo layout established	PASSED	Daluvuyo Magagane
AI Accountability Logged	Human verification, decisions, and hallucination fixes documented	PASSED	Daluvuyo Magagane

Baseline Readiness Statement

The CivicConnect engineering team has formally reviewed and verified the contents of PED v1.0. The requirements baseline, traceability matrix, risk register, forward engineering considerations, and GitHub governance policies are fully integrated, consistent, and defended. The project foundation is established under strict engineering controls, justifying progression to Milestone 2 (Architecture, Design & Engineering Decisions).